

# PAPER\_372 &#8212; Compressed UQFF with B/Bcrit Superconductivity

### Author: Daniel T. Murphy | Session 101 | Source: grokshare11254865.txt (lines 2700–3400)

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### Star Magic UQFF Whitepaper Series

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### Abstract

This paper presents the Compressed UQFF formulation, a multi-term master gravity equation that incorporates Newtonian gravity, Hubble expansion, superconductivity via the B/Bcrit flux-quenching factor, environmental coupling, cosmological constant contribution, quantum coherence, fluid dynamics, and dark matter perturbation. The framework is parameterised for seven astrophysical systems and validated via unit test against SGR1745-2900. This is the first UQFF formulation to explicitly incorporate Bardeen-Cooper-Schrieffer (BCS) superconductivity quenching into the gravitational coupling (Linear Meissner form; see PAPER\_375 for the exponential form).

### 1. Master Equation

$$\begin{aligned} g_{\mathrm{UQFF}}(r,t) = & \frac{G M(t)}{r^2} \cdot [1 + H(t,z)] \cdot \left[1 - \frac{B}{B_{\mathrm{crit}}}\right] \cdot [1 + F_{\mathrm{env}}] \\ & + \frac{\Lambda c^2}{3} \\ & + \frac{\hbar}{\Delta x \cdot \Delta p} \int (\psi_{\mathrm{total}})^* \hat{H} \psi_{\mathrm{total}} \, dV \cdot \frac{2\pi}{t_{\mathrm{Hubble}}} \\ & + \rho_{\mathrm{fluid}} V g \\ & + (M_{\mathrm{vis}} + M_{\mathrm{DM}}) \left( \frac{\Delta \rho}{\rho} + \frac{3GM}{r^3} \right) \end{aligned}$$

where  $H(t,z) = H_0 t$  (Newtonian cosmological expansion approximation),  $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$ .

### 2. Modular Component Functions

| Function             | Formula                   | Constants                                    |
|----------------------|---------------------------|----------------------------------------------|
| compressed_base      | $GM/r^2$                  | $G = 6.674 \times 10^{-11}$                  |
| compressed_expansion | $1 + H_0 t$               | $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$ |
| compressed_super_adj | $1 - B/B_{\mathrm{crit}}$ | linear Meissner                              |
| compressed_env       | 1.0                       | default                                      |

| Function                | Formula                                                         | Constants                                     |
|-------------------------|-----------------------------------------------------------------|-----------------------------------------------|
| compressed_cosm         | $\Lambda c^2/3$                                                 | $\Lambda = 1.1 \times 10^{-2} \text{ m}^{-2}$ |
| compressed_quantum      | $(\hbar/10^{-68}) \cdot 2.176 \times 10^{-18} \cdot (2\pi/t_H)$ | $t_H = 4.35 \times 10^1 \text{ s}$            |
| compressed_fluid        | $\rho_f v_{gl}$                                                 | from MUGESystem                               |
| compressed_perturbation | $(M+M_{DM})(\delta\rho/\rho + 3GM/r^3)$                         | $\delta\rho/\rho = 10^{-}$                    |

3. System Parameters

| System                   | M (kg)                 | r (m)               | B (T)               | Bcrit (T)          | Vsys (m³)              | ffluid (Hz)            |
|--------------------------|------------------------|---------------------|---------------------|--------------------|------------------------|------------------------|
| Magnetar SGR1745-2900    | $2.984 \times 10^{30}$ | $1 \times 10^$      | $1 \times 10^{10}$  | $1 \times 10^{11}$ | $4.189 \times 10^{12}$ | $1.269 \times 10^{-1}$ |
| Sagittarius A*           | $8.155 \times 10^3$    | $1 \times 10^{12}$  | $1 \times 10^{-}$   | $1 \times 10^{-}$  | $3.552 \times 10^$     | $3.465 \times 10^{-}$  |
| Tapestry Starbirth       | $1.989 \times 10^3$    | $3.086 \times 10^1$ | $1 \times 10^{-}$   | $1 \times 10^{-3}$ | $1 \times 10^3$        | $1 \times 10^{-12}$    |
| Westerlund 2             | $1.989 \times 10^3$    | $3.086 \times 10^1$ | $1 \times 10^{-}$   | $1 \times 10^{-3}$ | $1 \times 10^3$        | $1 \times 10^{-12}$    |
| Pillars of Creation      | $1.989 \times 10^{32}$ | $9.46 \times 10^1$  | $1 \times 10^{-}$   | $1 \times 10^{-3}$ | $3.552 \times 10^$     | $8.457 \times 10^{-1}$ |
| Rings of Relativity      | $1.989 \times 10^3$    | $3.086 \times 10^1$ | $1 \times 10^{-}$   | $1 \times 10^{-}$  | $1 \times 10^$         | $1 \times 10^{-}$      |
| Student's Guide Universe | $1 \times 10^3$        | $1 \times 10^2$     | $1 \times 10^{-10}$ | $1 \times 10^{-}$  | $1 \times 10^0$        | $1 \times 10^{-1}$     |

4. Validation

**Unit test:** test\_compute\_compressed\_MUGE(SGR1745-2900)

Expected:  $1.782 \times 10^3 \text{ m/s}^2$

(Dominated by compressed\_base × expansion; B/Bcrit = 0.1 → 90% retention)

5. Physical Interpretation

The  $[1 - B/B_{\mathrm{crit}}]$  factor models the Meissner effect: as the magnetic field B approaches the critical field Bcrit, gravitational coupling is quenched—mirroring how a Type-I superconductor expels magnetic flux below Bcrit. The compressed UQFF thus unifies gravitomagnetic quenching with cosmological expansion and quantum corrections in a single framework. (For Type-II exponential treatment, see PAPER\_375.)

6. Implementation

**C++:** STAR\_MAGIC\_09SEPT\_UQFF\_MODULE.cpp, namespace CompressedUQFF **Python:** CondensedPhysics4.py, class CompressedUQFFBcritSuperconductivityCalculator (CP4 #20) **WOLFRAM\_TERM:** WOLFRAM\_TERM\_COMPRESSED\_UQFF\_BCRIT

